RANDOM FOREST MACHINE LEARNING PROJECT

Dataset: California Housing Prices | **Task:** Regression

1. Problem Statement

The objective is to build a machine learning regression model that predicts the median house value of a California district using housing, population, income and location-related features.

2. Dataset

The project uses the public California Housing Prices dataset from Kaggle (20,640 rows and 10 columns). The target variable is *median_house_value*. The dataset includes numerical features and the categorical feature *ocean_proximity*.

Source: Kaggle – California Housing Prices

3. Data Preprocessing

The data was explored using summary statistics, missing-value checks and visualizations. The target column was separated from the input features, followed by an 80:20 train-test split. Missing numerical values were imputed using the median calculated from the training data. The categorical feature was converted into binary columns using One-Hot Encoding. The resulting feature matrices contained 13 columns.

4. Model and Evaluation

A Random Forest Regressor was trained. GridSearchCV with 3-fold cross-validation was used for hyperparameter tuning. The selected parameters were n_estimators=200, max_depth=None and min_samples_split=2. The tuned model was evaluated on the held-out test set.

Evaluation metric	Result
MAE	$31,486.80
MSE	2,383,729,507.63
RMSE	$48,823.45
R^2 score	0.8181
Best cross-validation R^2	0.8133

5. Key Findings and Conclusion

Median income was the most influential feature according to the model's feature-importance scores. The tuned model achieved an R^2 score of 0.8181 on the test set, indicating that it explains about 81.81% of the variation in target values in this test sample. It performed slightly better than the original model on the reported test metrics. These results show that Random Forest can model relationships in this dataset, although prediction errors remain.

Limitation: The dataset's median house values are capped at $500,001, so the model cannot distinguish values above that ceiling. Feature importance indicates model reliance, not causation.